

ARCEN, Netherlands

WMO: 063910

Lat: 51.497N

Lon: 6.196E

Elev: 62

StdP: 14.66

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.0	23.8	13.4	10.9	23.8	18.1	13.7	25.6	23.7	48.4	21.2	48.4	5.4	70	0.510

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	86.5	68.1	82.3	66.9	78.8	65.2	70.4	81.6	68.7	78.9	67.0	75.8	6.7	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.8	98.8	74.3	65.2	93.2	72.1	63.7	88.6	70.6	34.3	81.8	32.9	79.3	31.5	75.8	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.9	16.5	14.4	DB	15.4	94.1	5.7	4.2	11.3	97.1	8.0	99.6	4.8	102.0	0.7	105.0	(4)
(5)			WB	14.8	74.0	5.6	1.8	10.8	75.4	7.5	76.4	4.4	77.4	0.4	78.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.5	37.6	39.3	43.9	50.6	57.4	62.5	65.9	64.7	59.6	52.2	44.7	39.2	(6)
(7)		DBStd	11.91	7.59	7.08	6.52	6.83	6.43	5.87	5.90	5.14	5.26	6.32	6.31	7.53	(7)
(8)		HDD50	1550	385	304	206	75	11	0	0	0	1	50	178	340	(8)
(9)		HDD65	5126	848	721	653	433	248	115	59	67	177	396	609	800	(9)
(10)		CDD50	2114	2	3	18	94	240	376	493	455	290	119	19	5	(10)
(11)		CDD65	216	0	0	0	1	12	42	87	57	16	1	0	0	(11)
(12)		CDH74	2075	0	0	0	35	157	427	822	509	120	5	0	0	(12)
(13)		CDH80	608	0	0	0	3	26	117	276	163	23	0	0	0	(13)
(14)	Wind	WSAvg	6.8	8.2	8.0	7.5	6.7	6.6	6.1	6.0	5.5	5.7	6.7	6.9	8.0	(14)
(15)	Precipitation	PrecAvg	30.4	2.6	2.4	2.0	1.7	2.3	2.8	3.0	3.1	2.4	2.6	2.8	2.9	(15)
(16)		PrecMax	41.3	5.8	5.2	4.0	4.3	4.7	9.1	6.1	6.4	6.2	7.0	4.5	6.4	(16)
(17)		PrecMin	22.3	0.2	0.4	0.2	0.0	0.5	0.5	0.9	0.9	0.4	1.1	0.3	1.2	(17)
(18)		PrecStd	3.9	1.2	1.2	1.1	0.9	1.1	1.7	1.4	1.7	1.6	1.1	1.1	1.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.9	58.3	67.3	79.2	83.5	89.2	93.5	91.3	83.6	73.0	62.4	56.7	(19)
(20)			MCWB	51.8	49.9	53.8	59.3	66.1	69.1	69.4	69.6	66.8	62.9	57.6	52.9	(20)
(21)		2%	DB	52.9	54.0	61.8	72.7	78.1	83.3	86.9	84.5	77.7	67.1	58.5	54.5	(21)
(22)			MCWB	50.2	49.0	51.8	57.5	62.1	66.8	68.0	68.5	65.7	60.6	54.7	51.5	(22)
(23)		5%	DB	50.5	51.6	57.7	67.5	74.5	78.8	82.3	79.5	73.3	63.7	56.0	52.2	(23)
(24)			MCWB	47.9	47.7	50.3	55.3	60.5	65.1	67.1	66.4	63.8	58.7	52.9	49.6	(24)
(25)		10%	DB	47.9	49.2	54.0	63.0	70.5	74.8	78.4	75.3	69.3	61.1	53.7	49.6	(25)
(26)			MCWB	45.5	46.0	48.4	53.4	59.0	63.3	65.8	64.5	61.4	57.1	51.1	47.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.0	52.0	55.6	61.3	68.2	72.5	72.8	72.9	69.2	63.9	57.8	54.0	(27)
(28)			MCD B	54.5	55.3	63.3	73.2	79.4	84.8	85.8	86.3	78.7	70.4	60.7	55.6	(28)
(29)		2%	WB	50.3	50.1	53.2	58.6	64.9	69.0	70.3	69.9	66.8	61.2	55.4	51.9	(29)
(30)			MCD B	52.6	52.9	59.1	69.6	74.0	79.3	81.7	80.6	75.1	66.1	58.0	54.2	(30)
(31)		5%	WB	48.1	48.2	51.2	56.5	62.3	66.8	68.7	67.7	64.8	59.5	53.4	49.7	(31)
(32)			MCD B	50.4	51.1	56.4	65.3	70.8	75.6	79.3	77.1	71.8	63.2	55.7	51.8	(32)
(33)		10%	WB	45.7	46.1	49.2	54.4	60.0	64.7	66.9	65.9	62.8	57.6	51.1	47.5	(33)
(34)			MCD B	47.7	49.1	53.7	61.9	68.1	72.2	75.7	73.3	68.0	60.5	53.4	49.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.6	10.7	14.6	18.2	18.3	18.8	18.0	17.8	16.8	13.0	9.9	8.1	(35)
(36)			MCDBR	9.5	13.3	21.4	25.7	25.1	25.8	25.7	25.5	21.9	15.7	11.5	9.2	(36)
(37)			MCWBR	8.2	9.1	12.7	12.7	11.5	11.9	10.0	11.3	11.2	9.8	8.5	7.7	(37)
(38)		5% WB	MCD BR	9.6	11.8	18.1	22.2	21.5	21.8	22.0	22.3	19.8	14.7	10.7	8.9	(38)
(39)			MCWBR	8.6	8.8	11.4	11.8	11.2	11.3	9.6	10.9	10.7	9.7	8.6	7.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.330	0.353	0.383	0.418	0.423	0.421	0.430	0.422	0.395	0.367	0.344	0.326	(40)	
(41)		taud	2.378	2.344	2.266	2.201	2.229	2.250	2.228	2.276	2.342	2.403	2.384	2.352	(41)	
(42)		Ebn at Noon	214	241	255	259	264	265	260	256	249	233	206	193	(42)	
(43)		Edh at Noon	20	27	34	41	42	42	42	38	32	25	20	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	233	446	806	1297	1568	1688	1600	1362	997	587	276	178	(44)	
(45)		RadStd	33	86	120	144	138	144	199	124	109	51	34	23	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (86 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page